

Linear Algebra 2

→ Intro to vector

→ Distances

→ Matrix Multiplication

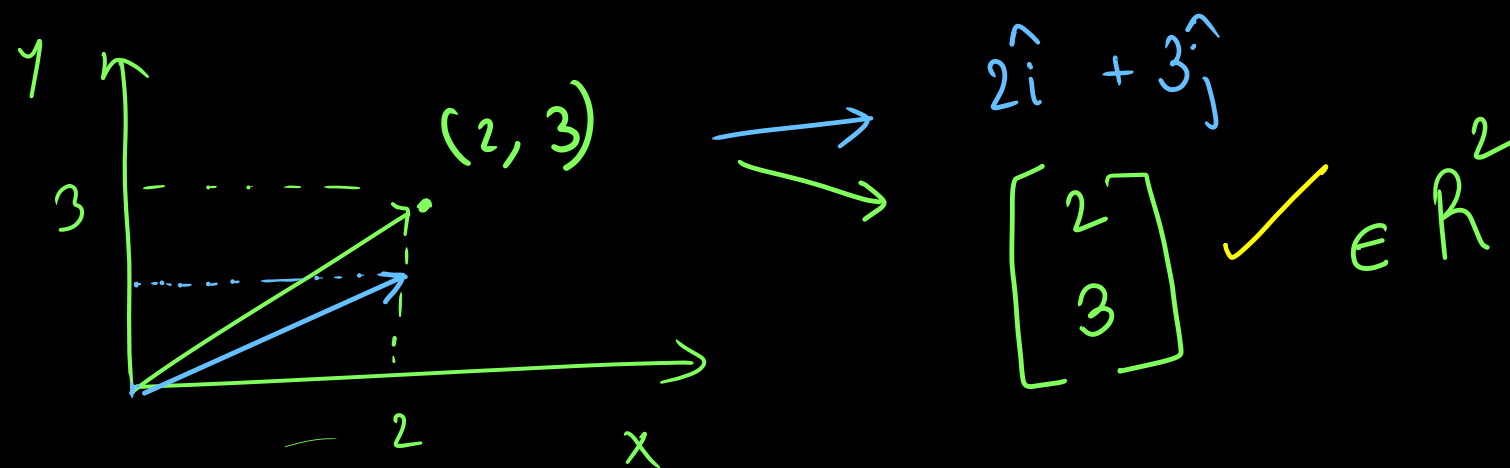
→ Angle b/w vector

→ Vector Projection

→ Vector

↳ What is a Vector?

↳ Magnitude + Direction

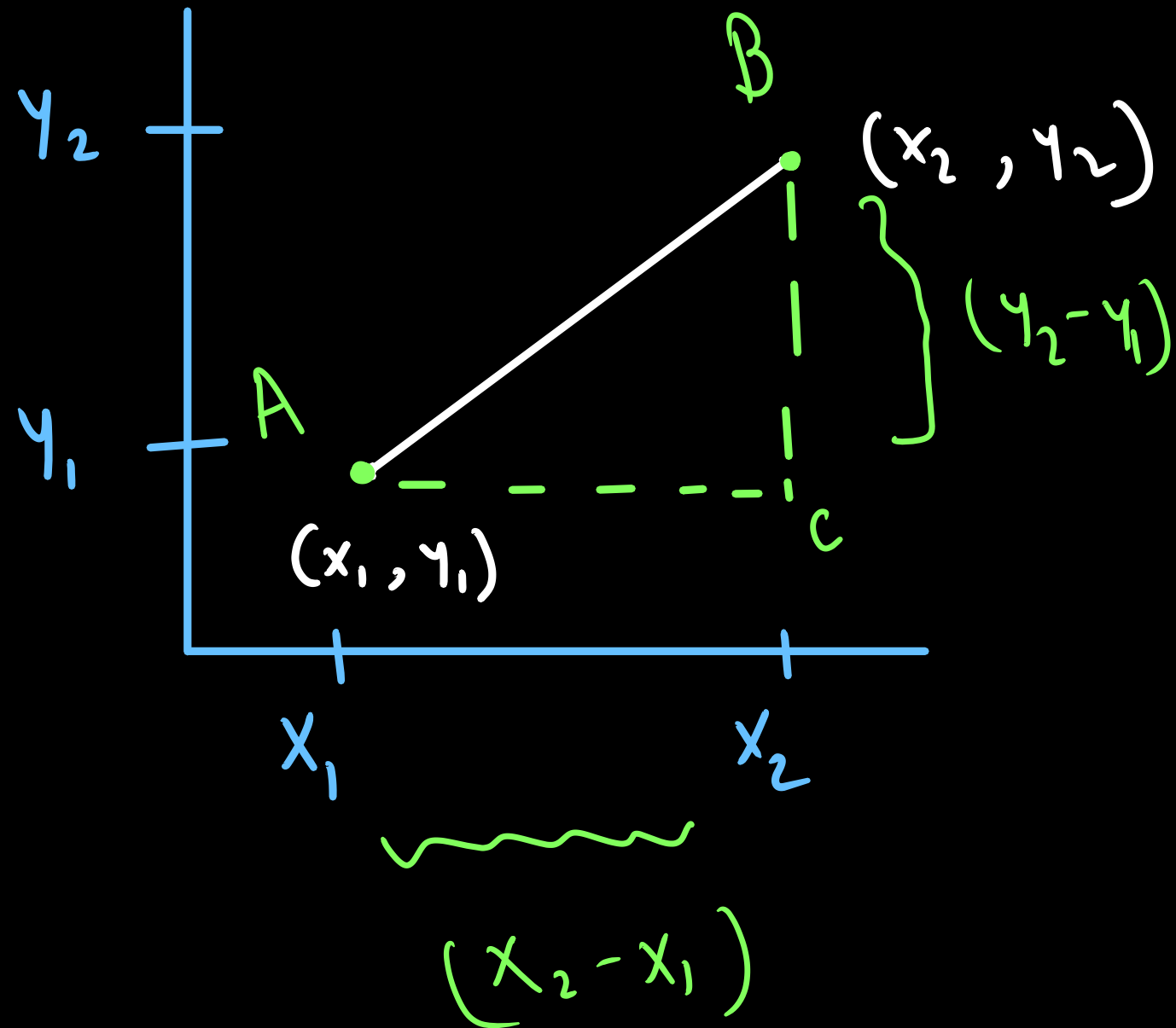


$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^d$$

$$\begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} \in \mathbb{R}^3$$

$$\begin{bmatrix} -2 \\ 10 \\ -5 \end{bmatrix} \in \mathbb{R}^3$$

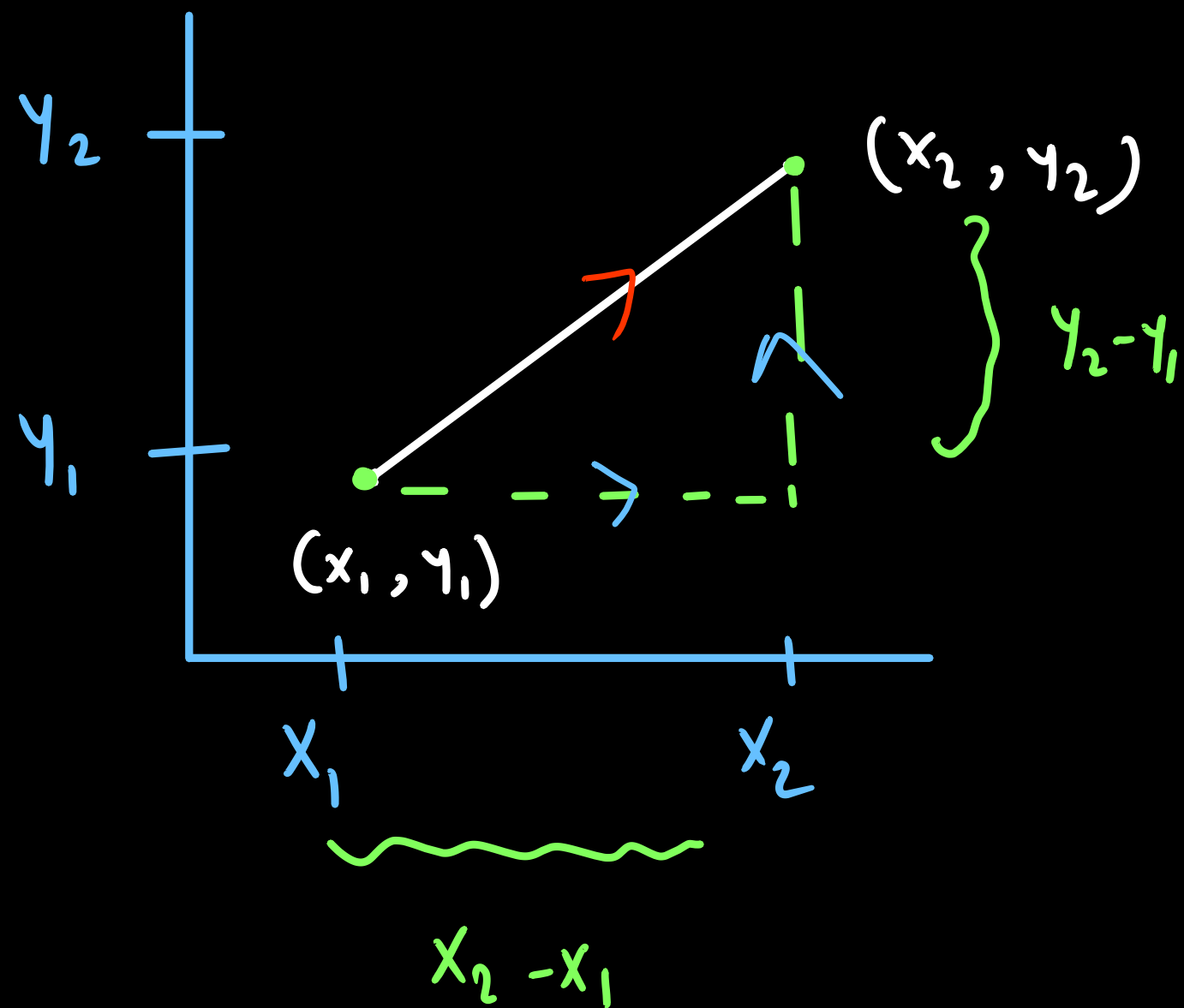
→ Distance formula



→ Euclidean dist

→ Manhattan dist

$$AB = \sqrt{AC^2 + CB^2}$$
$$= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$



$$\checkmark |5| = 5$$

$$\checkmark |-5| = 5$$

Euclidean

$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Manhattan

$$|x_2 - x_1| + |y_2 - y_1|$$

$$A = (1, 1)$$

$$B = (4, 3)$$

$$\begin{aligned} \text{Euclidean} &= \sqrt{(4-1)^2 + (3-1)^2} = \sqrt{3^2 + 2^2} \\ &= \sqrt{13} \end{aligned}$$

$$\begin{aligned} \text{Manhattan} &= |4-1| + |3-1| = |3| + |2| \\ &= 5 \end{aligned}$$

Magnitude of vector = Distance of Vector $\begin{bmatrix} x \\ y \end{bmatrix}$ from $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$

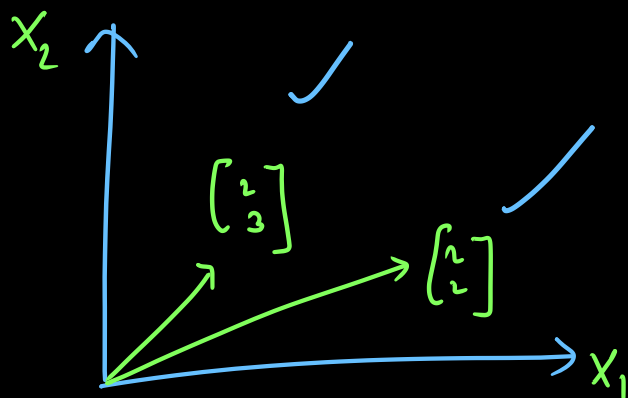
Euclidean
Magnitude $\rightarrow$

$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$(x_1, y_1) = (0, 0)$$

$$= \sqrt{2^2 + 3^2}$$

$$= \sqrt{13}$$



$$= \underbrace{\sqrt{x^2 + y^2}}$$

Manhattan Magnitude

$\rightarrow$

$$|2 - 0| + |3 - 0| = |2| + |3| = 5$$

$$\rightarrow \underbrace{|x - 0| + |y - 0|} = |x| + |y|$$

(x, y)

$(0, 0)$

Norm of a vector

↳ Magnitude of vector
(length)

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \quad \begin{array}{l} L_2 \text{ norm} = \text{Euclidean Magnitude} \\ L_1 \text{ norm} = \text{Manhattan Magnitude} \end{array}$$

→ Regularization
in ML

$$\begin{aligned} \|x\|_2 &= \sqrt{(-2)^2 + (3)^2} \\ &= \sqrt{13} \end{aligned}$$

$$\begin{aligned} \|x\|_1 &= |-2| + |3| \\ &= 2 + 3 = 5 \end{aligned}$$

Norm of a vector

L_2 = Euclidean Magnitude

$$\|x\|_2 = \sqrt{x_1^2 + x_2^2 + \dots + x_d^2}$$

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix}$$

$\|x\|_1 = L_1$ = Manhattan Magnitude

$$= |x_1| + |x_2| + \dots + |x_d|$$

→ What is norm of $\begin{bmatrix} -2 \\ 3 \end{bmatrix}$?

$$L_2 \rightarrow \sqrt{(-2)^2 + (3)^2} = \sqrt{13}$$

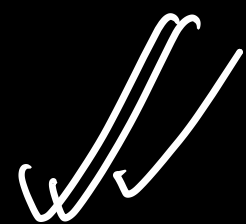
$$L_1 \rightarrow |-2| + |3| = |5| = 5$$

Matrix Multiplication

$$\underbrace{C}_{m \times n} = \underbrace{A}_{m \times d} \underbrace{B}_{d \times n}$$

$$\boxed{d = k}$$

$$\underline{\underline{d \neq k}}$$



$\rightarrow$ No of col of 1st mat
 $=$ No of row of 2nd mat

$\rightarrow$ No of row of 1st matrix
 $=$ No of row of o/p

$\rightarrow$ No of col of 2nd matrix
 $=$ No of col of o/p

$$\begin{bmatrix} \underline{1} & \underline{2} \\ \underline{3} & \underline{4} \end{bmatrix} \begin{bmatrix} \underline{5} & \underline{6} \\ \underline{7} & \underline{8} \end{bmatrix}$$

$$= \begin{bmatrix} 1 \times 5 + 2 \times 7 & 1 \times 6 + 2 \times 8 \\ 3 \times 5 + 4 \times 7 & 3 \times 6 + 4 \times 8 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

$$1 \times \textcircled{2} \quad \textcircled{2} \times 1$$

$$1 \times 1$$

$$\begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

2×1

$$\begin{bmatrix} 1 & 2 \end{bmatrix}$$

1×2

$$\textcircled{2 \times 2}$$

→ Dot product

$$A^T B = C_{1 \times 1}$$

$1 \times 2 \quad 2 \times 1$

$$A = 2\hat{i} + 3\hat{j}$$

$$= \begin{bmatrix} 2 \\ 3 \end{bmatrix}_{2 \times 1}$$

$$A B^T = C_{2 \times 2}$$

$2 \times 1 \quad 1 \times 2$

$$B = 4\hat{i} + 5\hat{j}$$

$$= \begin{bmatrix} 4 \\ 5 \end{bmatrix}_{2 \times 1}$$

$$\begin{bmatrix} - \\ - \end{bmatrix}$$

$$\underline{A \cdot B} = 2 \times 4 + 3 \times 5$$

$$= 8 + 15$$

$$= 23$$

$$\underline{\underline{A^T B}} = \begin{bmatrix} 2 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \end{bmatrix}$$

$1 \times 2 \quad 2 \times 1$

$$\underline{\underline{1 \times 1}}$$

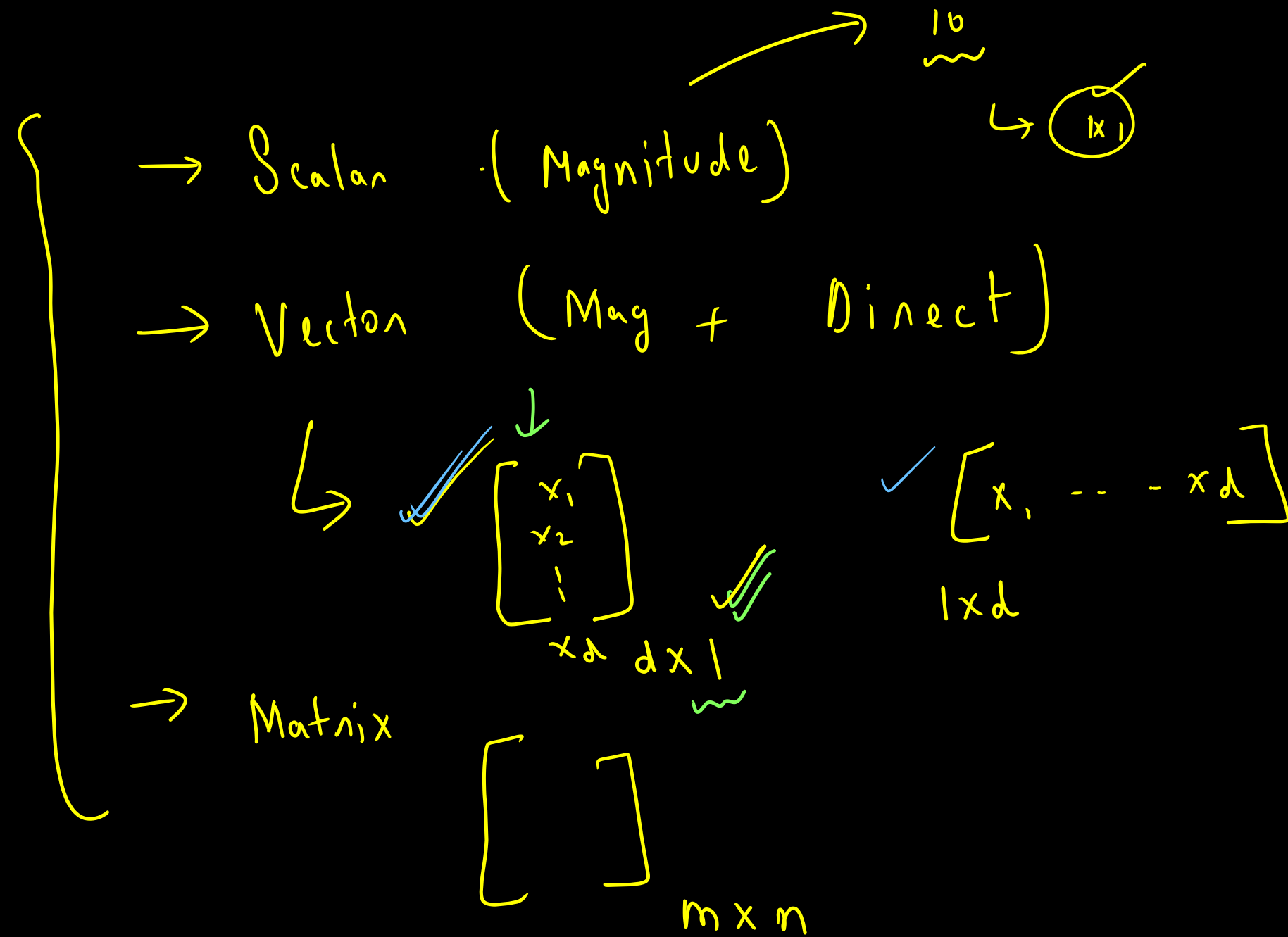
$$= \begin{bmatrix} 2 \times 4 + 3 \times 5 \end{bmatrix}$$

$$A = \begin{bmatrix} 2 \\ 3 \end{bmatrix}_{2 \times 1}$$

$$A^T = \begin{bmatrix} 2 & 3 \end{bmatrix}_{1 \times 2}$$

$$= \underline{\underline{23}}$$

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \quad A^T = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$$

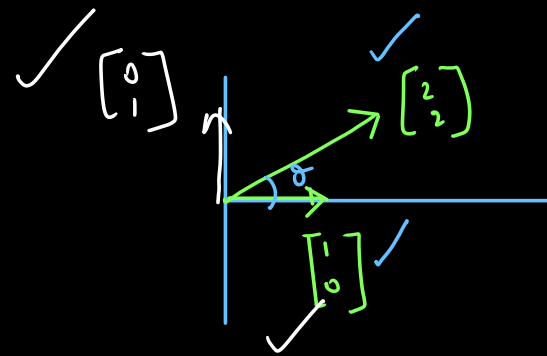


$$A = \begin{bmatrix} \end{bmatrix}_{d \times 1}$$

$$B = \begin{bmatrix} \end{bmatrix}_{d \times 1}$$

Angle b/w 2 vectors

$$\begin{bmatrix} 2 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 2 \times 1 + 2 \times 0$$



$$\cos \theta = \frac{x^T y}{\|x\| \|y\|}$$

$$= \frac{\text{Dot Prod}}{\text{Mag}}$$

$$-1 \leq \cos \theta \leq 1$$

$$\cos \theta = \frac{2 \times 1 + 2 \times 0}{\sqrt{2^2 + 2^2} \sqrt{1^2 + 0^2}} = \frac{2}{\sqrt{8} \sqrt{1}} = \frac{2}{2\sqrt{2}} = \frac{1}{\sqrt{2}}$$

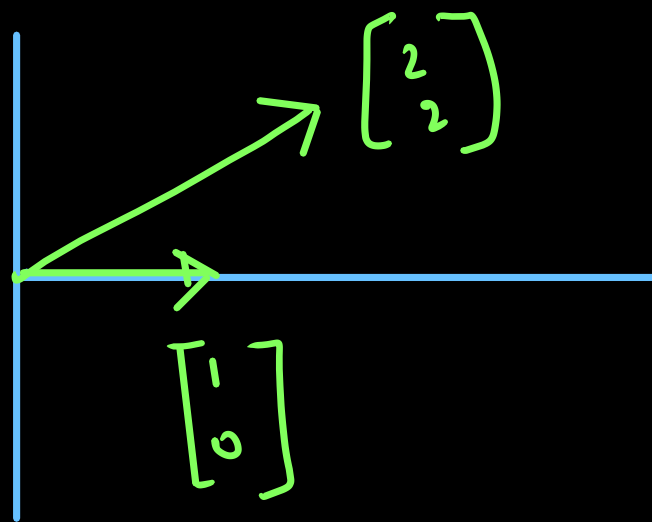
$$\cos \theta = \frac{1}{\sqrt{2}}$$

$$\theta = \cos^{-1}\left(\frac{1}{\sqrt{2}}\right) = 45^\circ$$

$$= \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 0 + 0 = 0$$

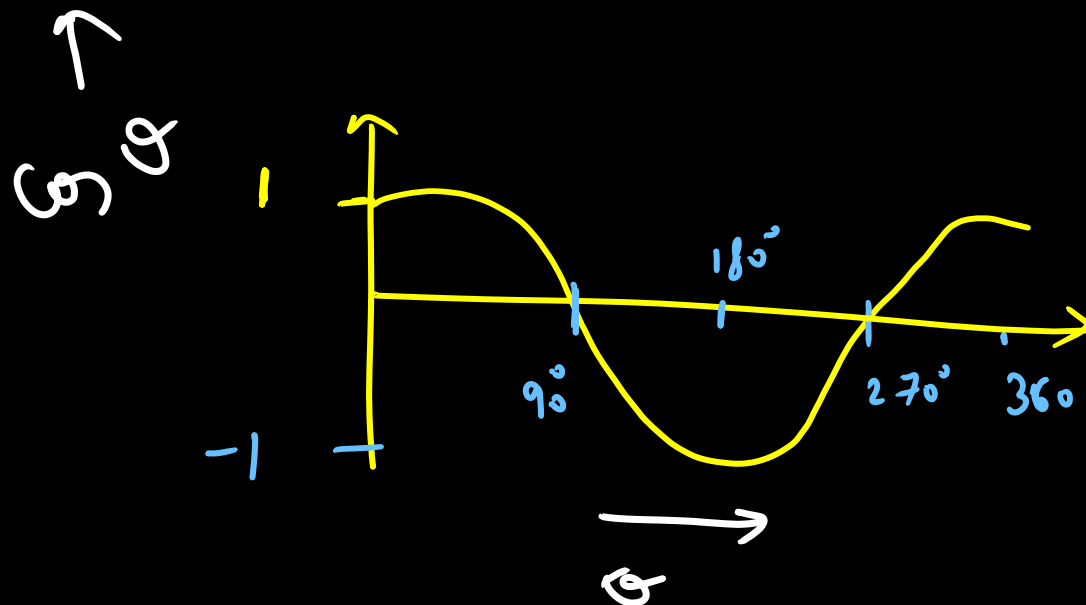
$$\cos \theta = 0$$

$$\theta = \cos^{-1}(0) = 90^\circ$$



$$\cos \theta = \frac{x^T y}{\|x\| \|y\|}$$

$$\underline{\underline{-1 \leq \cos \theta \leq 1}}$$



$$\begin{aligned} 0 < \theta < 90^\circ &\longrightarrow +ve \\ 270^\circ < \theta < 360^\circ &\longrightarrow +ve \\ 90^\circ < \theta < 270^\circ &\longrightarrow -ve \end{aligned}$$

→ Break until 22:25 PM

Coordinate Geometry & Linear Algebra

$$w_1 x_1 + w_2 x_2 + \dots + w_d x_d + w_0 = 0$$

$$\checkmark \quad w = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_d \end{bmatrix} \quad d \times 1$$

$$\checkmark \quad x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix} \quad d \times 1$$

$$\underline{w^T x} + w_0 = 0 \rightarrow b$$

$$\checkmark \quad \underbrace{[w_1 \ w_2 \ \dots \ w_d]}_{w^T} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix}}_x$$

$$= w_1 x_1 + w_2 x_2 + \dots + w_d x_d$$

$$[w_1 \ w_2 \ \dots \ w_d] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix}$$

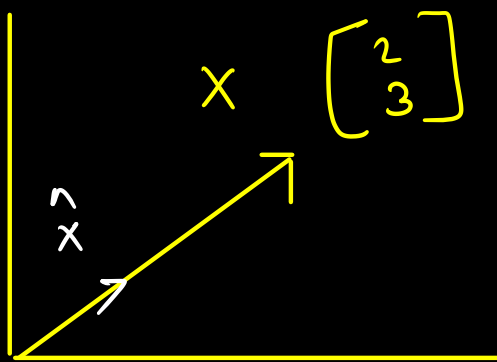
→ Unit vector

↳ Magnitude 1 & direction

Convert any vector to
unit vector

$$\|x\| = \sqrt{13}$$

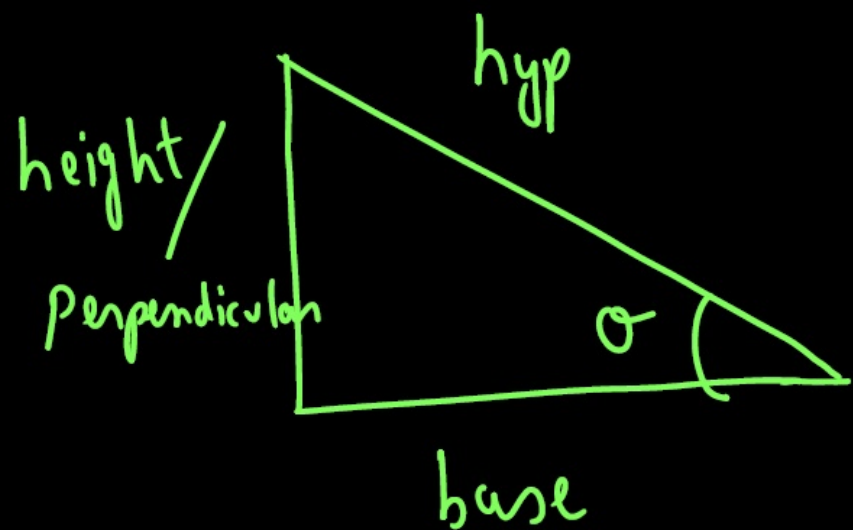
$$\checkmark \hat{x} = \frac{x}{\|x\|} = \begin{bmatrix} \frac{2}{\sqrt{13}} \\ \frac{3}{\sqrt{13}} \end{bmatrix}$$



$$\|\hat{x}\| = \sqrt{\left(\frac{2}{\sqrt{13}}\right)^2 + \left(\frac{3}{\sqrt{13}}\right)^2} = \sqrt{\frac{4}{13} + \frac{9}{13}} = \sqrt{\frac{13}{13}}$$

$$= 1$$

Trigonometry

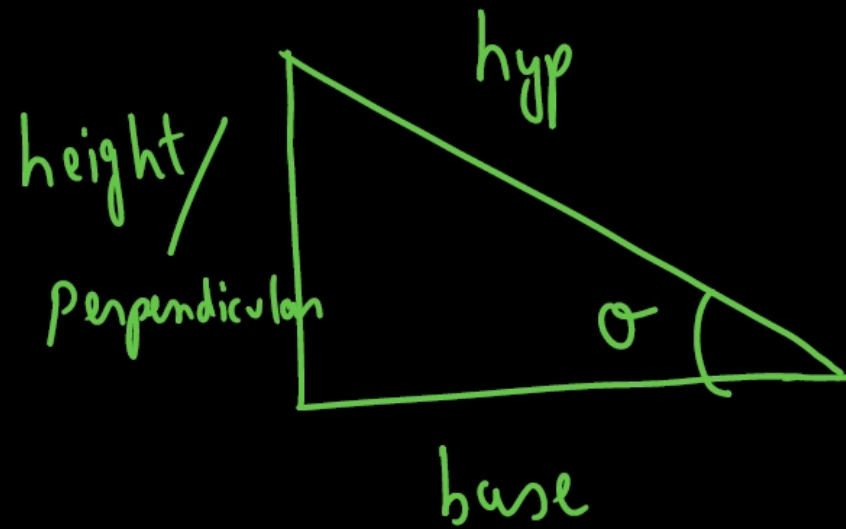


$$\sin \theta = \frac{\text{height}}{\text{hyp}}$$

$$\cos \theta = \frac{\text{base}}{\text{hyp}}$$

$$\tan \theta = \frac{\text{height}}{\text{base}}$$

Trigonometry



$$\sin \theta = \frac{\text{height}}{\text{hyp}}$$

$$\cos \theta = \frac{\text{base}}{\text{hyp}}$$

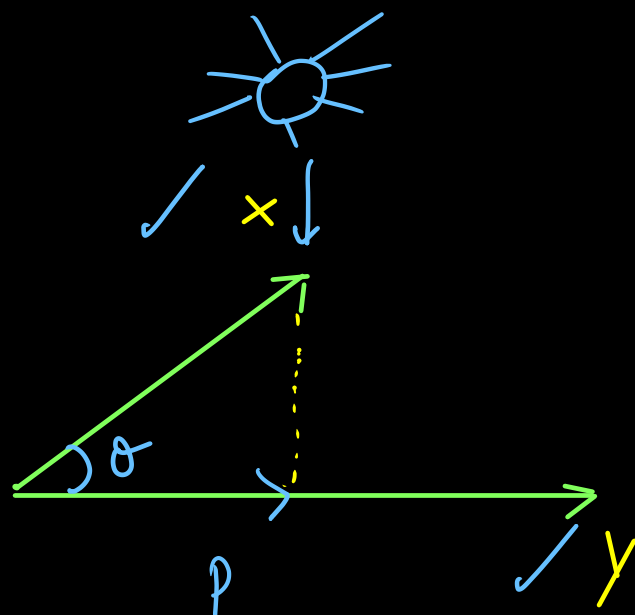
$$\tan \theta = \frac{\text{height}}{\text{base}}$$

Projection of a Vector

$$\cos \theta = \frac{\|p\|}{\|x\|} \rightarrow \textcircled{1}$$

$$\cos \theta = \frac{x^T y}{\|x\| \|y\|} \rightarrow \textcircled{2}$$

← Angle b/w x & y



Scalar Projection

$$\frac{\|p\|}{\|x\|} = \frac{x^T y}{\|x\| \|y\|}$$

$$\|p\| = \frac{x^T y}{\|y\|}$$

Vector Projection

→ Scalar Proj * $\hat{y}$

$$\frac{x^T y}{\|y\|} * \frac{y}{\|y\|}$$

~~$\|y\|$~~